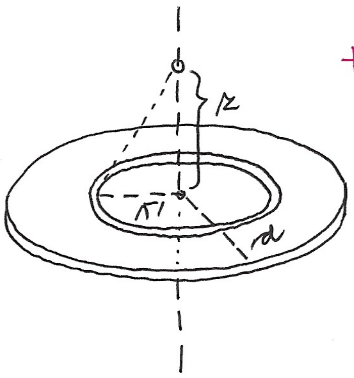
EMP, 1. A PISNI IZPIT

1 NABITI OKROGLI PLOŠČI

a) $\underbrace{\text{– ena plošča}}$



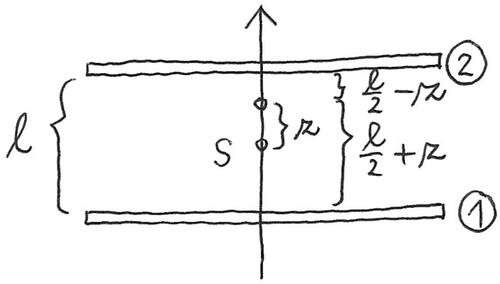
+(1) $\left\{ de = \sigma \cdot 2\pi r' \, dr' \right.$

+(2) $\left\{ dU = \dfrac{de}{4\pi\varepsilon_0 \sqrt{r'^2+z^2}} = \dfrac{\sigma}{4\varepsilon_0} \dfrac{2r'dr'}{\sqrt{r'^2+z^2}} \right.$

+(3) $$\left\{ \begin{aligned} U &= \frac{\sigma}{4\varepsilon_0} \int_0^a \frac{2r'dr'}{\sqrt{r'^2+z^2}} = \frac{\sigma}{4\varepsilon_0} 2\sqrt{r'^2+z^2}\Big|_0^a \\ &\underbrace{U = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{z^2+a^2} - |z|\right)} \rightarrow \underbrace{U(\infty) = 0}, \end{aligned} \right.$$

kar je v redu!

$\underbrace{\text{– obe plošči}}$



$$\left. \begin{aligned} U_1 &= \frac{\sigma}{2\varepsilon_0}\left(\sqrt{\left(\frac{l}{2}+z\right)^2+a^2} - \frac{l}{2} - z\right) \\ U_2 &= \frac{\sigma}{2\varepsilon_0}\left(\sqrt{\left(\frac{l}{2}-z\right)^2+a^2} - \frac{l}{2} + z\right) \end{aligned} \right\} +(4)$$

$$U(z) = U_1 + U_2$$

$$\boxed{U(z) = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{\left(\frac{l}{2}+z\right)^2+a^2} + \sqrt{\left(\frac{l}{2}-z\right)^2+a^2} - l\right)} \Bigg\} +(5)$$

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b) +(6) $$\left\{ \begin{aligned} &\sqrt{\left(\frac{l}{2}\pm z\right)^2+a^2} = \left(\frac{l^2}{4} + a^2 \pm lz + z^2\right)^{\frac{1}{2}} = \sqrt{a^2+\frac{l^2}{4}}\left(1 + \frac{z^2 \pm lz}{a^2+\frac{l^2}{4}}\right)^{\frac{1}{2}} \approx \\ &\approx \sqrt{a^2+\frac{l^2}{4}} + \frac{1}{2\sqrt{a^2+\frac{l^2}{4}}}\left(z^2 \pm lz\right) \\ &\underbrace{U(z) \approx \frac{\sigma}{2\varepsilon_0}\left[\left(2\sqrt{a^2+\frac{l^2}{4}} - l\right) + \frac{1}{\sqrt{a^2+\frac{l^2}{4}}} z^2\right]} \end{aligned} \right.$$

, saj se linearna člena odštejeta

$$U(z) \approx U_0 + V z^2$$

$$\downarrow$$

+(7) $$\left\{ \begin{aligned} &U(x,y,z) \approx U_0 + V \underbrace{\left(z^2 - \frac{1}{2}x^2 - \frac{1}{2}y^2\right)}, \\ &r^2 = x^2 + y^2 \\ &\boxed{U(r,z) = \frac{\sigma}{2\varepsilon_0}\left[\left(2\sqrt{a^2+\frac{l^2}{4}} - l\right) + \frac{1}{\sqrt{a^2+\frac{l^2}{4}}}\left(z^2 - \frac{1}{2}r^2\right)\right]} \end{aligned} \right.$$

da bo $\underbrace{\nabla^2 U = 0}$, x & y pa morata nastopati SIMETRIČNO c)

$$E_r = -\frac{\partial V}{\partial r} = \boxed{\frac{\sigma}{2\varepsilon_0}\,\frac{1}{\sqrt{a^2+\frac{\ell^2}{4}}}\,r}$$

$$E_z = -\frac{\partial V}{\partial z} = \boxed{-\frac{\sigma}{2\varepsilon_0}\,\frac{1}{\sqrt{a^2+\frac{\ell^2}{4}}}\,2z}$$

+(8)

– zadnji rezultat lahko dobimo tudi neposredno iz rezultata pod a)

$$E_z = -\frac{\partial V}{\partial z} = -\frac{\sigma}{2\varepsilon_0}\left[\frac{2\left(\frac{\ell}{2}+z\right)}{2\sqrt{\left(\frac{\ell}{2}+z\right)^2+a^2}} + \frac{-2\left(\frac{\ell}{2}-z\right)}{2\sqrt{\left(\frac{\ell}{2}-z\right)^2+a^2}}\right] =$$

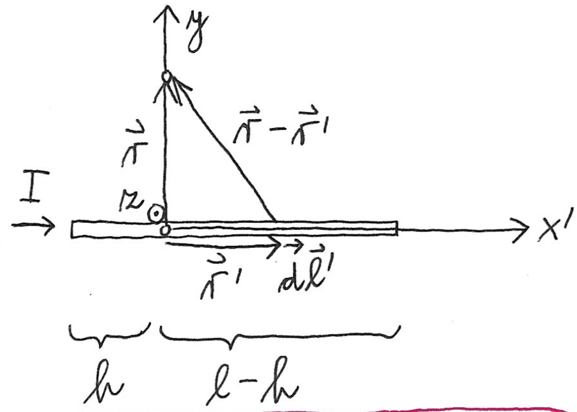
$$= -\frac{\sigma}{2\varepsilon_0}\left[\frac{2z}{2\sqrt{\left(\frac{\ell}{2}+z\right)^2+a^2}} + \frac{2z}{2\sqrt{\left(\frac{\ell}{2}-z\right)^2+a^2}}\right] \xrightarrow{z \ll \ell} \underline{-\frac{\sigma}{2\varepsilon_0}\,\frac{2z}{\sqrt{a^2+\frac{\ell^2}{4}}}}$$

8 + → $\boxed{1}$

$\boxed{2}$ IZBOČENI VODNIK

a) - poljuben ravni odsek

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}' \times (\vec{r}-\vec{r}')}{|\vec{r}-\vec{r}'|^3}$$



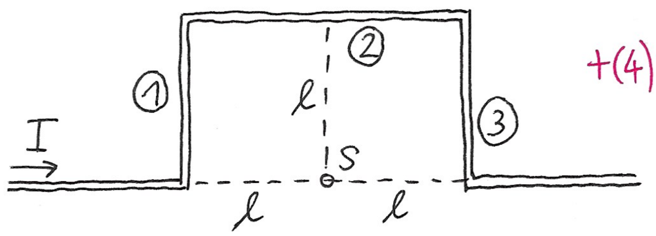
+(1)
$$\vec{r} = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}, \quad \vec{r}' = \begin{bmatrix} x' \\ 0 \\ 0 \end{bmatrix}, \quad d\vec{l}' = \begin{bmatrix} dx' \\ 0 \\ 0 \end{bmatrix}$$

$$|\vec{r}-\vec{r}'| = \sqrt{r^2 + x'^2}$$

$$d\vec{l}' \times (\vec{r}-\vec{r}') = r\,dx'\,\hat{e}_z$$

+(2)
$$\vec{B} = \frac{\mu_0 I}{4\pi} r\,\hat{e}_z \int_{-h}^{l-h} \frac{dx'}{(x'^2+r^2)^{\frac{3}{2}}} = \frac{\mu_0 I}{4\pi} r\,\hat{e}_z \left. \frac{x'}{r^2\sqrt{x'^2+r^2}} \right|_{-h}^{l-h}$$
+(3)

$$\underbrace{\vec{B} = \frac{\mu_0 I}{4\pi r} \left[\frac{l-h}{\sqrt{(l-h)^2+r^2}} + \frac{h}{\sqrt{h^2+r^2}} \right] \hat{e}_z}$$



+(4) ① $h = 0$, $r = l$ $\quad [\;] = \frac{l}{\sqrt{l^2+l^2}} = \frac{1}{\sqrt{2}}$

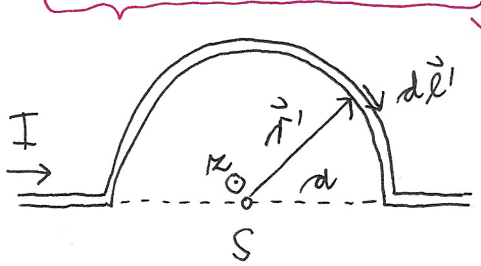
+(5) ② $l \to 2l$, $h = l$, $r = l$ $\quad [\;] = 2 \cdot \frac{l}{\sqrt{l^2+l^2}} = \frac{2}{\sqrt{2}}$

③ = ①

$$\vec{B}_{\sqcap} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 = \frac{\mu_0 I}{4\pi l} \left(\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) (-\hat{e}_z)$$

$$\boxed{B_{\sqcap} = \frac{\mu_0 I}{\sqrt{2}\pi l}}$$

b) $\underbrace{4l = \pi a \Rightarrow \frac{l}{a} = \frac{\pi}{4}}$



$$\vec{B}_{\Omega} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}' \times (\vec{r}-\vec{r}')}{|\vec{r}-\vec{r}'|^3} = -\frac{\mu_0 I}{4\pi} \int \frac{d\vec{l}' \times \vec{r}'}{r'^3}$$

$r' = a$, $\quad d\vec{l}' \times \vec{r}' = a\,d\varphi \cdot a\,\hat{e}_z$ +(6)

$$\vec{B}_{\Omega} = -\frac{\mu_0 I}{4\pi} \hat{e}_z \frac{a^2}{a^3} \int_0^{\pi} d\varphi = \frac{\mu_0 I}{4a} (-\hat{e}_z)$$
+(7)

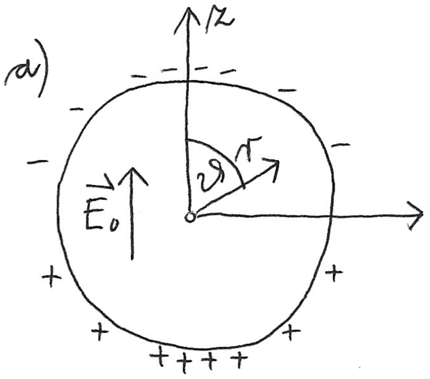
$$B_{\Omega} = \frac{\mu_0 I}{4a} = \boxed{\frac{\pi \mu_0 I}{16 l}}$$

+(8)
$$\frac{B_{\Omega}}{B_{\sqcap}} = \frac{\pi}{16}\sqrt{2}\pi = \boxed{\frac{\pi^2\sqrt{2}}{16} = 0.87}$$

8+ → $\boxed{1}$

3 NABITA KROGELNA LUPINA

a)



$$\boxed{U(r<a,\vartheta) = -E_0 r\cos\vartheta} \quad \text{podano} \Big\} +(1)$$

$$\downarrow \quad P_1(\cos\vartheta) \quad \uparrow$$

$$\Downarrow$$

$$U(r>a,\vartheta) = \frac{B}{r^2}\cos\vartheta \Big\} +(2)$$

RP1 ZVEZNOST potenciala pri $r=a$

$$+(3)\Big\{ -E_0 a\cos\vartheta = \frac{B}{a^2}\cos\vartheta \Rightarrow \underline{B=-E_0 a^3} \rightarrow \boxed{U(r>a,\vartheta) = -\frac{E_0 a^3}{r^2}\cos\vartheta}$$

RP2 GAUSSOV IZREK na površini, pri $r=a$

+(4)

$$de = \varepsilon_0 dS\left(E_r^{ZUN} - E_r^{NOT}\right)\Big|_\partial \Rightarrow \sigma = \frac{de}{dS} = \varepsilon_0\left(E_r^{ZUN}-E_r^{NOT}\right)\Big|_\partial$$

$$E_r^{ZUN}(a,\vartheta) = -\frac{2E_0 a^3}{r^3}\cos\vartheta\Big|_{r=a} = -2E_0\cos\vartheta$$

$$E_r^{NOT}(a,\vartheta) = E_0\cos\vartheta \quad \Rightarrow \quad \underline{\sigma = -3\varepsilon_0 E_0\cos\vartheta}$$

$$p_e = \int z\,\sigma\, dS = \int_{-1}^{1} a\cos\vartheta\left(-3\varepsilon_0 E_0\cos\vartheta\right) a^2\cdot 2\pi\, d(\cos\vartheta) =$$

$$= -6\pi\varepsilon_0 E_0 a^3\int_{-1}^{1}\cos^2\vartheta\, d(\cos\vartheta) = \boxed{-4\pi\varepsilon_0 E_0 a^3}$$

$$\Uparrow$$

ALI PREPROSTO $\quad U(r>a,\vartheta) = \dfrac{p_e\cos\vartheta}{4\pi\varepsilon_0 r^2} \Rightarrow \underline{\dfrac{p_e}{4\pi\varepsilon_0} = -E_0 a^3} \Big\} +(4)$

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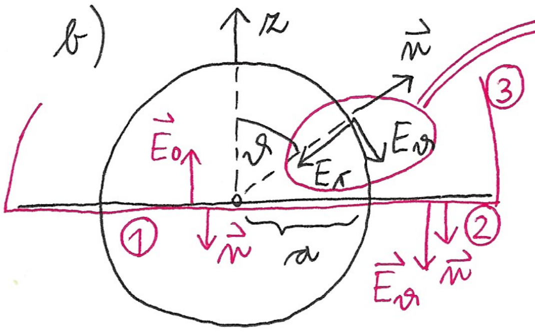
b)



integracijska ploskev : VELIKA polsfera

① & ② $\quad \vec{E} \parallel \vec{n} \Rightarrow \vec{E}(\vec{E}\cdot\vec{n}) - \frac{1}{2}E^2\vec{n} = \frac{1}{2}E^2\vec{n}$

$$① \quad \vec{F}_{e1} = \frac{\varepsilon_0}{2}\int E_0^2\vec{n}\,dS = \frac{\varepsilon_0}{2}E_0^2\left(-\hat{e}_z\right)\cdot \pi a^2$$

$$\underline{\vec{F}_{e1} = -\frac{1}{2}\pi\varepsilon_0 E_0^2 a^2\,\hat{e}_z} \Big\} +(5)$$

$$② \quad \vec{F}_{e2} = \frac{\varepsilon_0}{2}\int E_\vartheta^2\,\vec{n}\,dS \quad , \quad +(6)\Big\{ E_\vartheta = -\frac{1}{r}\frac{\partial U}{\partial\vartheta}\Big|_{\vartheta=\frac{\pi}{2}} = -\frac{E_0 a^3}{r^3}\sin\frac{\pi}{2} = -\frac{E_0 a^3}{r^3}$$

$$\underbrace{\vec{F}_{e2} = -\frac{\varepsilon_0}{2}E_0^2 a^6\,\hat{e}_z \underbrace{\int_a^\infty \frac{1}{r^6}\cdot 2\pi r\,dr}_{2\pi\cdot\frac{1}{4}\cdot\frac{1}{a^4}}}_{+(7)} = \underline{-\frac{1}{4}\pi\varepsilon_0 E_0^2 a^2\,\hat{e}_z} \Rightarrow \boxed{F_e = \frac{3\pi}{4}\varepsilon_0 E_0^2 a^2}$$

$3+\rightarrow\boxed{1}$ $\quad$ ③ $\vec{F}_{e3} = 0$ $\quad$ +(8)